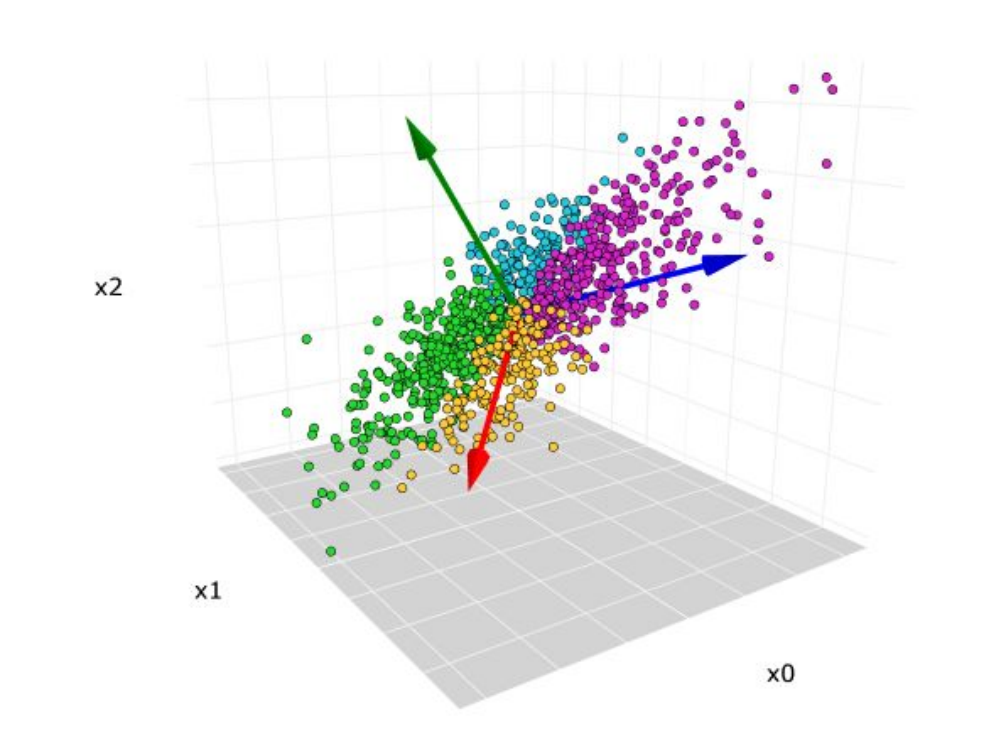
Principal Component Analysis



What is Principal Component Analysis (PCA)?

In practice, real data is messy, it has too many details, columns, and measurements so how do we see the big picture without getting lost?

Principal Component Analysis (PCA) is a technique used to simplify data by reducing the number of features while preserving as much important information as possible.

- PCA is an unsupervised learning technique
- It does not use labels or outcomes
- Instead, it finds patterns based only on the structure of the data

Basically PCA transforms data into a smaller set of variables that still capture most of the original information.

Why do we need PCA?

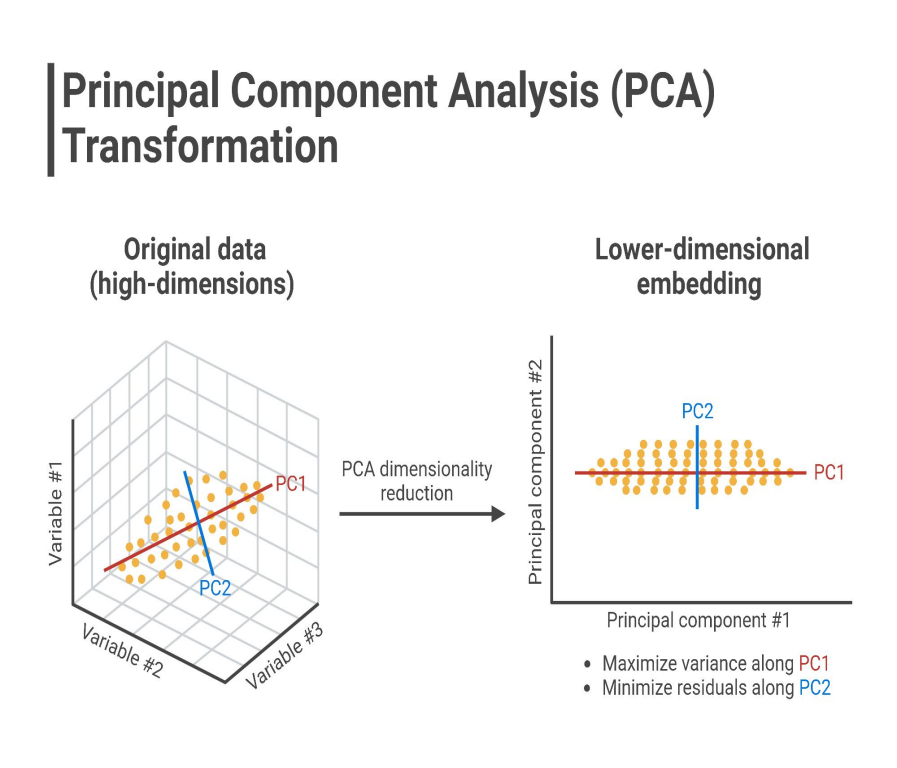
- High-dimensional data is hard to visualize and understand
- Many features are correlated or redundant
- Too many features can slow down models and increase complexity
- Noise in data can reduce model performance
- Visualization breaks beyond 3D

How PCA Works

- It works by first standardizing the data and then identifying directions (called principal components) that capture the maximum variation in the dataset. These components are new axes that represent the data in a more informative way, with the first component capturing the most variance and each subsequent component capturing the remaining variation while being independent of the previous ones.

- The data is then projected onto these principal components, allowing it to be represented using fewer dimensions. By selecting only the top components, PCA reduces the number of features while retaining most of the important information in the dataset.

- Because PCA focuses on maximizing variance and removing redundancy, it is effective for simplifying data, improving visualization, and reducing noise, especially when features are highly correlated.



Key Concepts / Intuition

Key Ideas:

- **Principal components = new axes capturing most variation**
- **PC1 captures the most, PC2 the next most (perpendicular), etc.**
- **Variance = information → more spread = more important**
- **Correlated features are redundant → PCA removes redundancy**

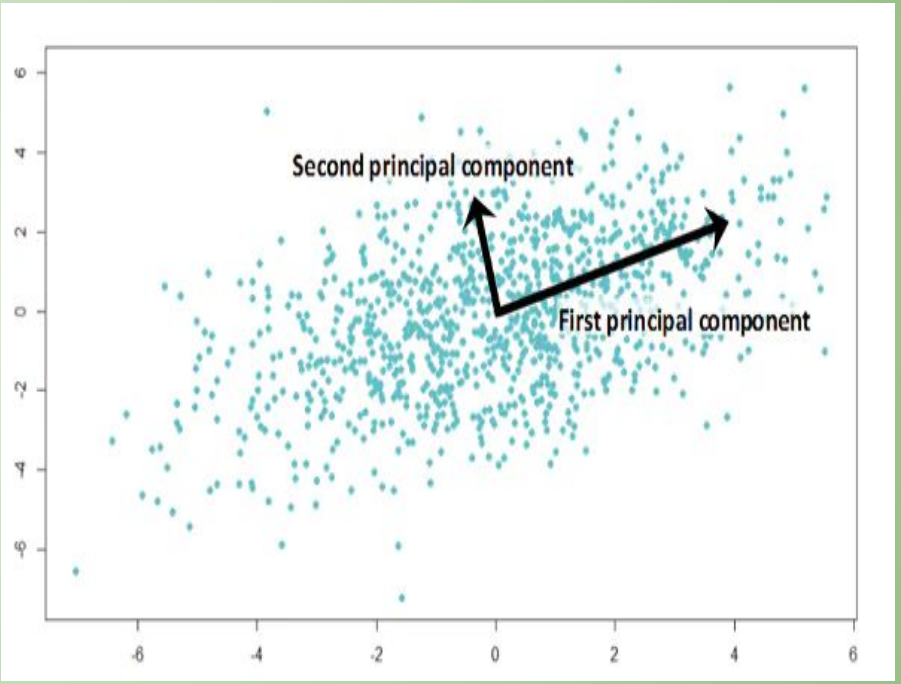
Takeaway: PCA re-expresses your data along the directions that matter most

Explained Variance / Component Selection

Key Ideas:

- Each component explains a % of total variance
- Higher variance = more important
- Use scree plot (look for elbow) or cumulative
- variance (e.g., keep 95%) to select components

Takeaway:
Keep the top components that capture most of the data's information



Interactive PCA Visualization

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Applications & Limitations of PCA

Where PCA is Used:

- Data visualization – reduce dimensions for plotting (2D/3D)
- Noise reduction – remove irrelevant or redundant features
- Preprocessing for ML models – speeds up training, improves performance
- Image compression – represent images with fewer components
- High-dimensional datasets – finance, genomics, sensors, etc.

Limitations:

- Only captures linear relationships
- Sensitive to scaling and outliers (must standardize data)
- Components can be hard to interpret and is not ideal for complex nonlinear patterns